another, with the wormhole configuration drawn so as to emphasize that it lies outside of ordinary space. In Figure 15.5b, we show the time evolution of this wormhole, assuming both its mouths are kept stationary. (The time slices are those of a stationary observer.) In Figure 15.5c, we show what happens when one wormhole mouth is loaded onto a spaceship and taken on a round-trip journey. Time for the moving mouth, just like time on a moving clock, slows down, so that the moving mouth is transported to the future. (If an hour elapses on a moving clock but a thousand years elapse on stationary clocks, the moving clock will have jumped into the stationary clocks' future.) Thus, instead of the stationary wormhole mouth's connecting, via the wormhole tunnel, to a mouth on the same time slice, it connects to a mouth on a *future* time slice, as in Figure 15.5c. Unless the wormhole mouths are moved further, the time difference between them will remain locked in. At any moment, should you enter one mouth and exit the other, you will have become a time traveler.

Building a Wormhole Time Machine

One blueprint for building a time machine is now clear. Step 1: find or create a wormhole wide enough for you, or anything you want to send through time, to pass. Step 2: establish a time difference between the wormhole mouths—say, by moving one relative to the other. That's it. In principle.